

Relazione del progetto annuale: CM-CNC

OBIETTIVO PROGETTO:

- L'obiettivo del progetto è quello di creare una macchina CNC a 3 assi (x,y,z), in grado di incidere le PCB e con la possibilità di cambiare la punta nel mezzo del processo di incisione, ritagliare il contorno o anche scavare i fori dei pin passanti;
- Il progetto comprende una parte software, in modo da convertire i file dei programmi in file comprensibili dalla macchina (gerber).

POTENZIALI DIFFICOLTÀ:

- Gestione del programma di conversione dal progetto della PCB (su Ultiboard o altri programmi), ai movimenti del trapano per inciderla;
- Problemi hardware nella costruzione di una struttura solida e funzionale;
- Utilizzo di una parte di struttura preesistente (vecchia stampante 3D, modificarla quasi interamente porterà via sicuramente tempo e risorse);
- Creazione di una GUI e gestione del display touch Nextion.

PROBLEMI RISCONTRATI:

- Lo schermo Nextion, dopo diversi utilizzi, ha smesso di funzionare inaspettatamente. Si è dovuto contattare il venditore per farlo sostituire. Questo inconveniente ha rallentato la riuscita del progetto. Si è quindi dovuto modificare il Gantt allungando i tempi.
- Si è deciso di optare per cambiare il potenziometro analogico a slitta per la calibrazione della punta, con un semplice ma preciso pulsante digitale, dopo la constatazione della poca precisione del potenziometro.

MIGLIORAMENTI OTTENUTI:

Movimento dei 3 assi in funzione, con il sistema di finecorsa funzionante.
Utilizzo di una funzione sviluppata per il controllo dei motori, consistente in: aumentare gradualmente la velocità per poi arrivare a regime e successivamente diminuirla gradualmente fino al fermarsi del motore, il tutto programmato per ogni distanza richiesta;
Creazione della PCB e gestione dello schermo Nextion tramite il software dedicato.

TABELLA PIN-OUT CNC:

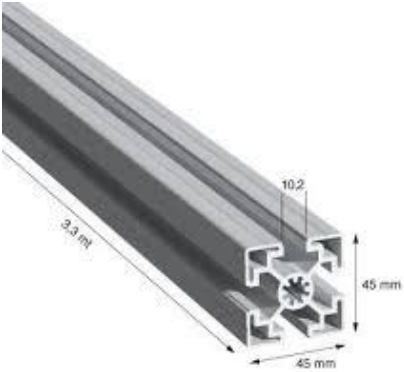
[Excel pin-out](#)

TABA		TABB	
PIN PRIMO ESP32	COLLEGAMENTO	PIN SECONDO ESP32	COLLEGAMENTO
36 (SP)	-	4	LIV-PUNTA
34	12 VOLT	5	-
16	DISPLAY-RX1	13	ENABLE
17	DISPLAY-TX1	14	STEP-Z2
21	ESP-FALG-1	16	DIR-X
33	ESP-FLAG-2	17	STEP-X
32	ESP-FLAG-3	18	ESP-FLAG-3
22	ESP-I2C-SCL	19	ESP-FLAG-2
23	ESP-I2C-SDA	21	ESP-FALG-1
26	LASER	22	ESP-I2C-SCL
13	LUCI-1	23	ESP-I2C-SDA
27	LUCI-2	25	DIR-Z1
25	PWM-MOT	26	STEP-Z1
35	SCHERMO	27	DIR-Z2
5	SD-CS	32	DIR-Y
19	SD-MISO	33	STEP-Y
14	SD-MOSI	34	FCX
18	SD-SCK	35	FC-Y
39 (SN)	SONDA TEMPERATURA	36 (SP)	FC-Z2
4	VENTOLE	39 (SN)	FC-Z1

LISTA MATERIALI NECESSARI CNC:

- 3x profili in alluminio;
- 4x nema 17;
- 4x driver TB6600 per nema17;
- 2x esp32;
- 3x relè digitale;
- 1x alimentatore atx;
- 1x striscia led bianchi;
- 1x modulo laser;
- 1x kit punte;
- 1x display touch per esp32;
- 4x barre filettate(30/40 cm);
- 6x barre (30/40 cm);
- 8x cuscinetti lineari (tipo 1);
- 4x cuscinetti lineari (tipo 2);
- 4x finecorsa ottici;
- 4x cuscinetti a sfere;
- 4x giunti (5mm, 8mm);
- 1x pulsante;
- 1x motore fresa;
- 1x lettore di carte SD;
- 1x sonda di temperatura (10k Ω);
- 1x amp. op. (LM358N).

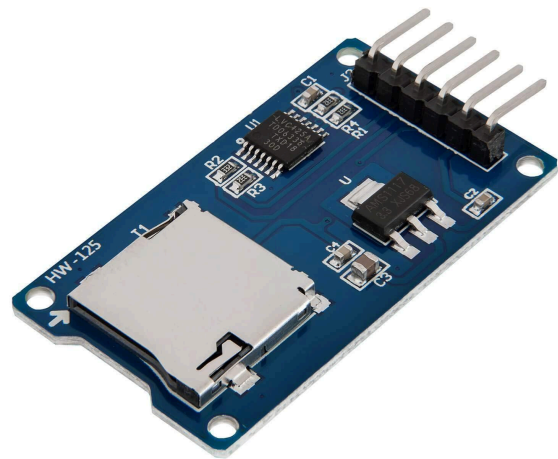
TABELLA MATERIALI CNC:

Profili in alluminio x3	 Technical drawing of an aluminum profile. The profile is shown in a perspective view with dimensions: 10.2 mm (width), 45 mm (height), and 45 mm (depth). The profile has a T-slot design.
Motori NEMA 17 2.5A x4	 A photograph of a NEMA 17 2.5A stepper motor. The motor has a silver-colored metal housing and a black rubber boot covering the motor body. The shaft is visible on the front.
Driver TB6600 x4	 A photograph of a TB6600 stepper motor driver module. The module is black with a green terminal block on the right side. It features various pins and components for interfacing with a microcontroller and a stepper motor.
Amp. Op. (LM358N) x1	 A photograph of an LM358N operational amplifier. The component is a small, black, dual in-line package (DIP) with eight pins. The text "24WTV19F2" and "LM358N" is visible on the top of the package.

Esp32 38pin x2



Lettore di carte SD x1



Alimentatore ATX x1



Striscia led x1



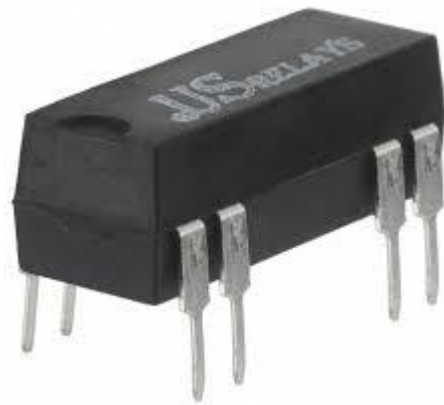
**Tappi per profilati in alluminio
x10**



**Angolare in alluminio per
profilati x30**



Relè digitali x3



Sonda di temperatura x1



Modulo laser x1



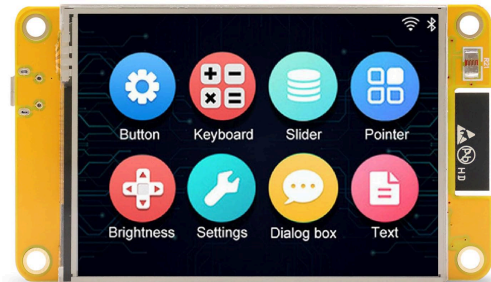
Kit punte x1



**Kit: motore fresa +
alimentatore x1**



Display touch per esp32 x1



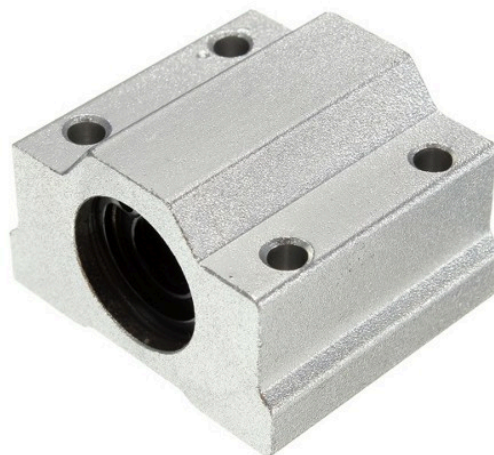
**Barre filettate (30/40 cm)
passo=2 mm $\varnothing$ =8 mm x4**



**Barre (30/40 cm) passo=2mm
 $\varnothing$ =8 mm x6**



**Cuscinetti lineari (tipo 1)
x8**



**Cuscinetti lineari (tipo 2)
x4**



Interruttore finecorsa x4



Boccole in ottone x4



Cuscinetti a sfere x4



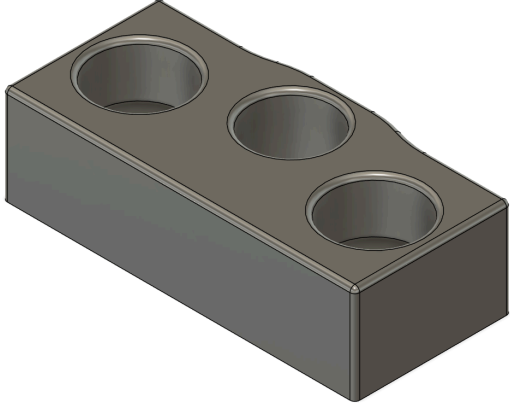
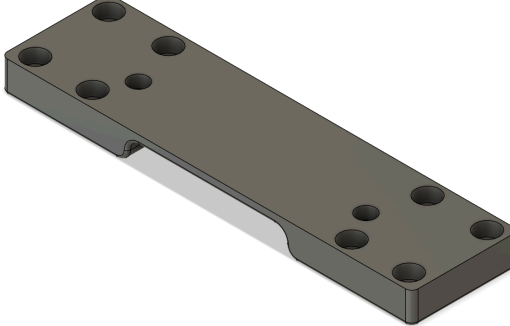
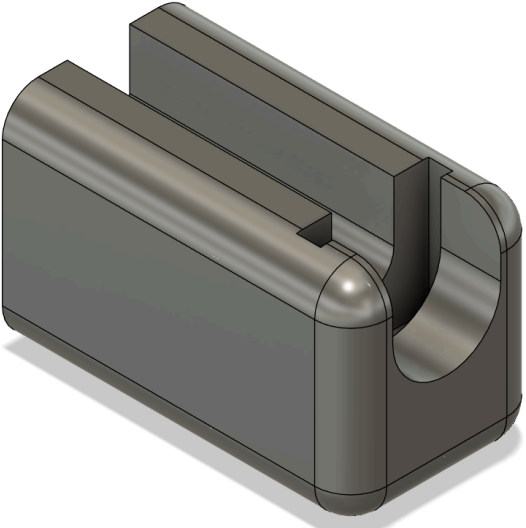
Giunti (5mm, 8mm) x4



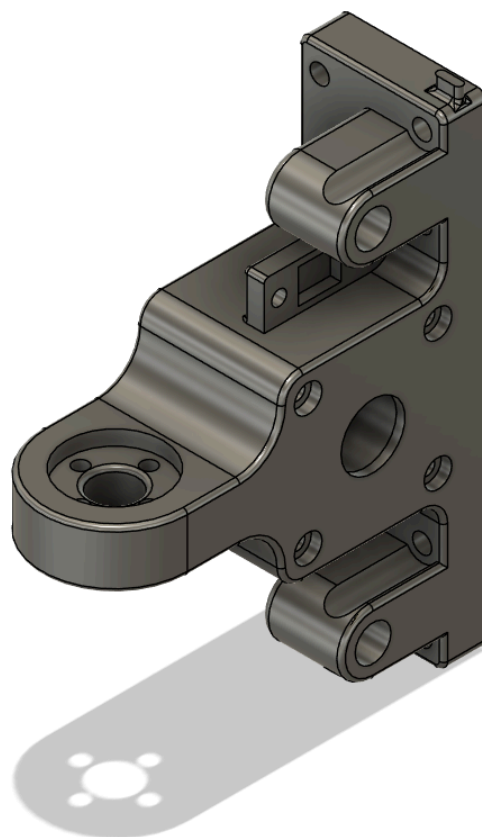
Pulsante x1



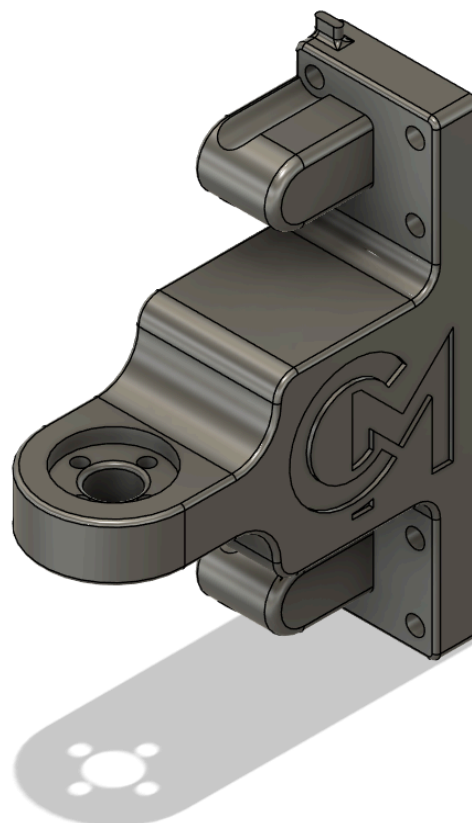
Stampe 3D:

Supporto Telaio-Barre (8 mm)	 A 3D rendered model of a dark gray rectangular block. On its top surface, there are three circular holes arranged in a slightly curved line. The block has a simple, functional design with sharp edges.
Supporto Cuscinetti (tipo 1) - Piano (X)	 A 3D rendered model of a long, thin, dark gray rectangular plate. It features two rows of small circular holes. The top row has four holes, and the bottom row has five holes. The plate has a slight thickness and a flat top surface.
Supporto Telaio - Barre (8 mm)	 A 3D rendered model of a dark gray support bracket. It has a complex shape with a large U-shaped cutout on one side. The bracket features rounded corners and a vertical flange on the back. It appears to be designed to hold or support a horizontal bar.

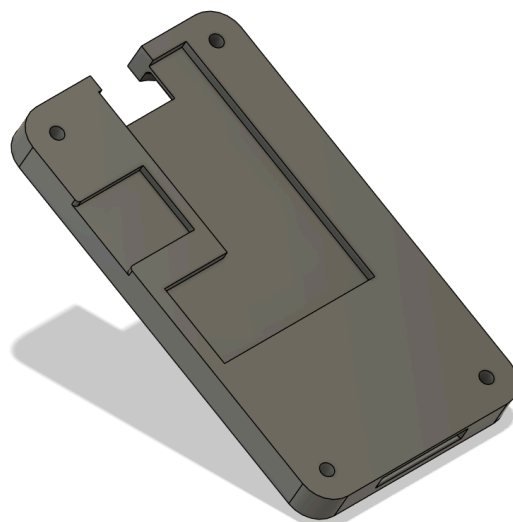
Supporto
Motore (Y) - Cuscinetti (tipo 1) -
Barre (8 mm) - Boccola (Z)



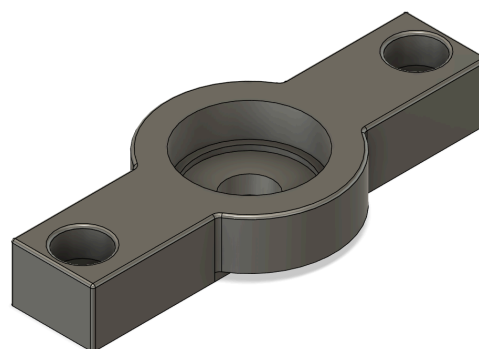
Supporto
Cuscinetti (tipo 1) -
Barre (8 mm) - Boccola (Z)



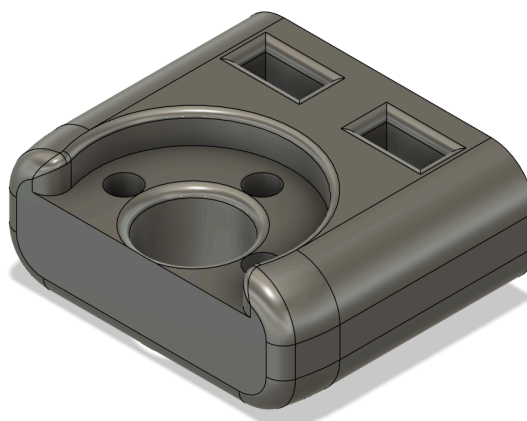
**Supporto
Telaio - Schermo**



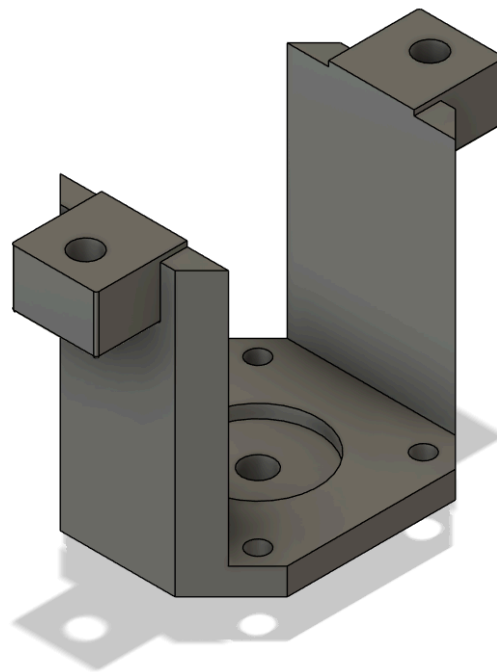
**Supporto
Cuscinetto - Telaio**



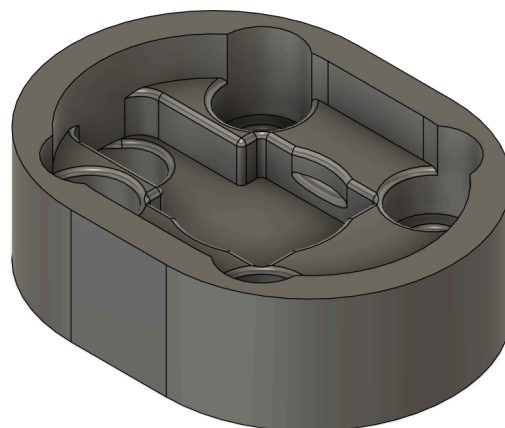
**Supporto
Boccola (X) - Piano (X)**



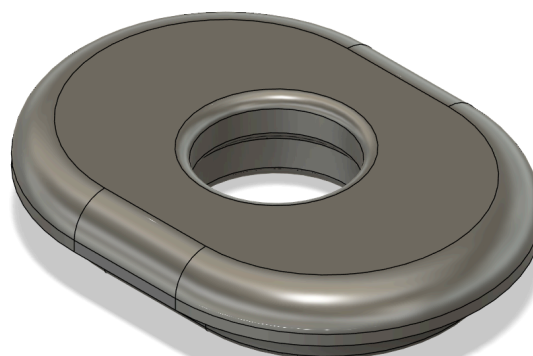
**Supporti x2
Motore (Z) - Telaio**



**Supporto
Telaio - Pulsante**



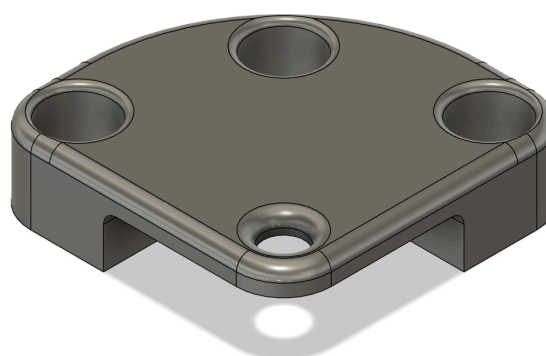
**Supporto
Telaio - Pulsante**



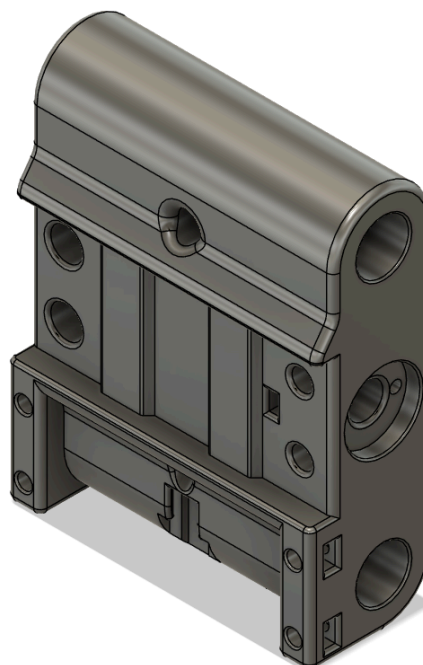
**Supporto
Telaio - Piatto**



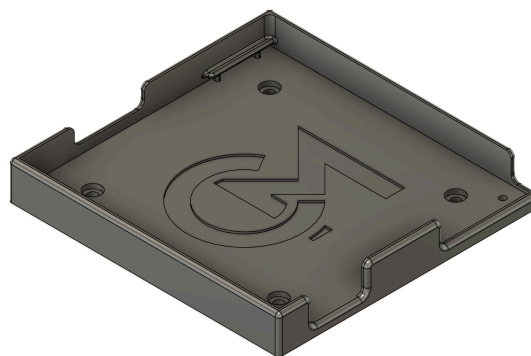
**Supporto
Telaio - Piatto**



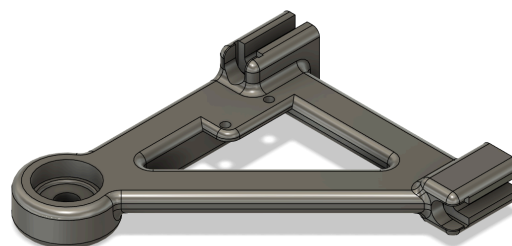
**Supporto
Telaio - Mandrino**



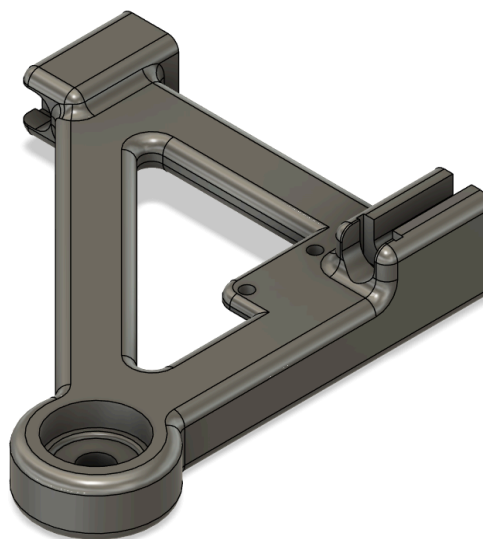
**Supporto
Alimentatori - Base**



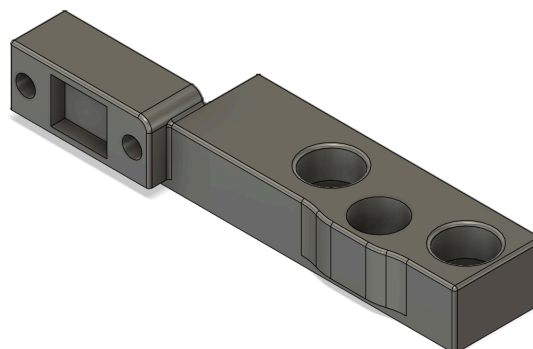
**Supporto
Laterale - Telaio - Sinistro**



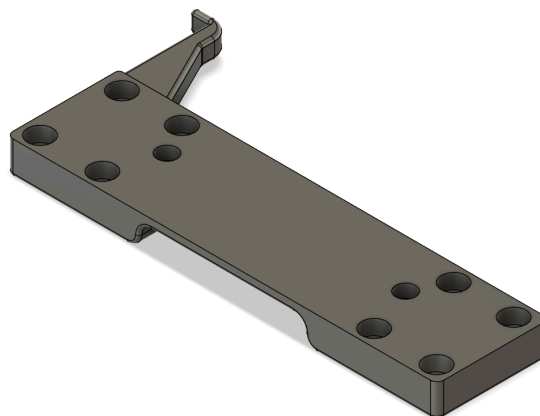
**Supporto
Laterale - Telaio - Destro**



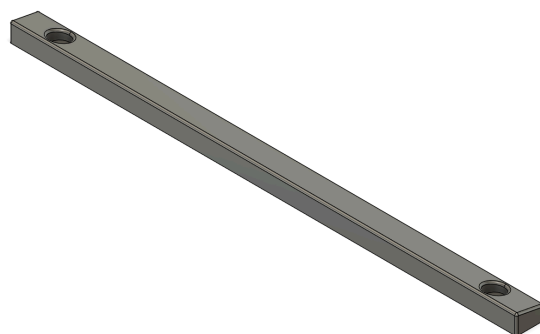
**Supporto
Telaio - Barre (8 mm) -
Finecorsa**



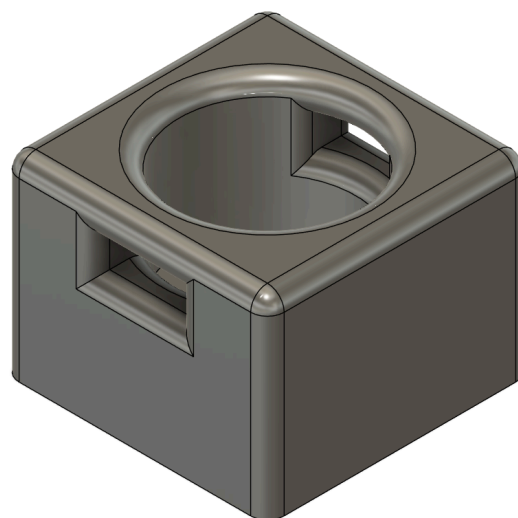
**Supporto
Cuscinetti (tipo 1) - Piano (X)**



**Supporto
Telaio - LED 12v**



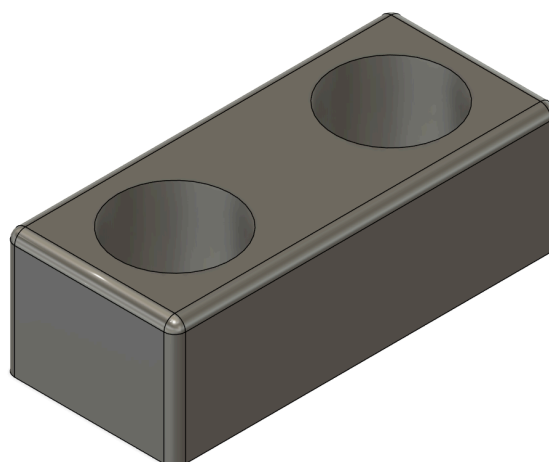
**Supporto
Telaio - Cavi**



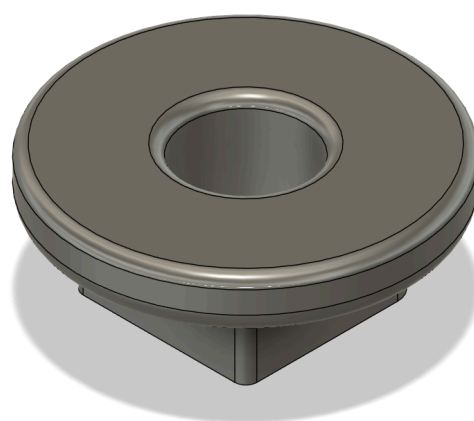
**Supporto
Telaio - Schermo**

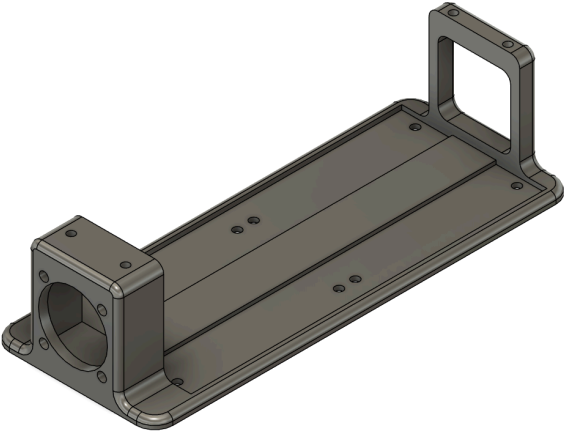
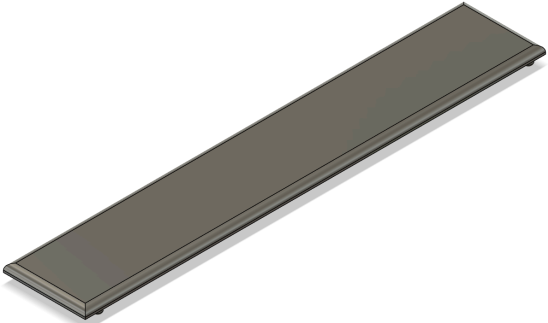
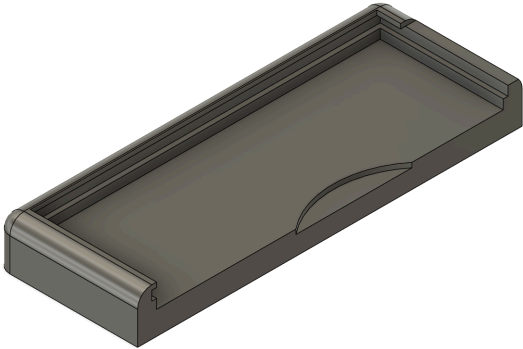


**Supporto
Telaio - Momentaneo**

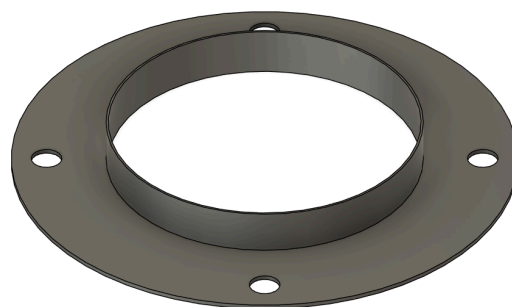


**Piedino
Telaio - Feltrino**

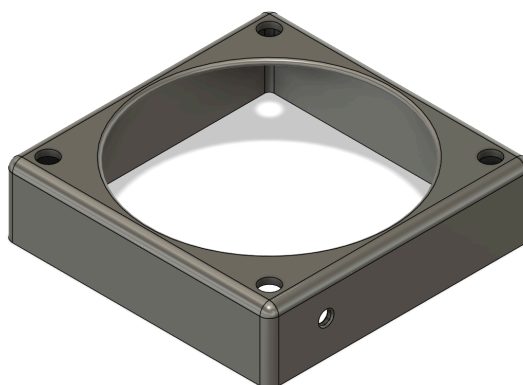


Supporto Driver - Ventola - Base	
Supporto Coperchio - Driver	
Supporto Base - PCB	
Supporto Base - PCB	

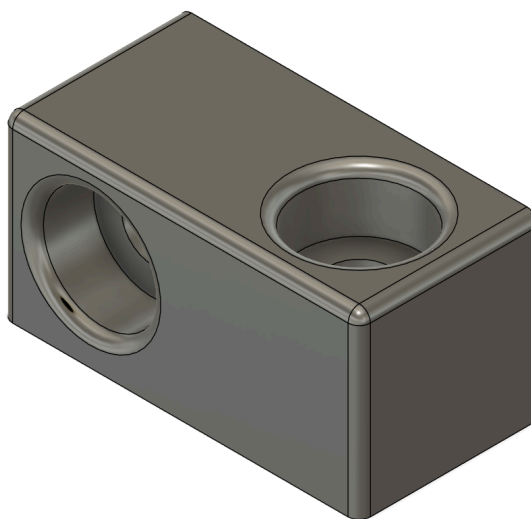
**Coperchio
Telaio - Ventole**



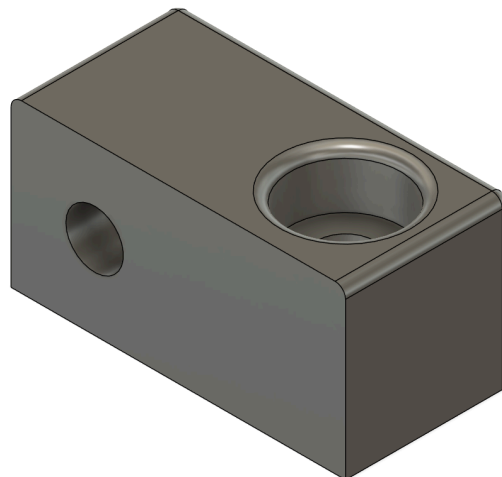
**Supporto
Telaio - Ventole**



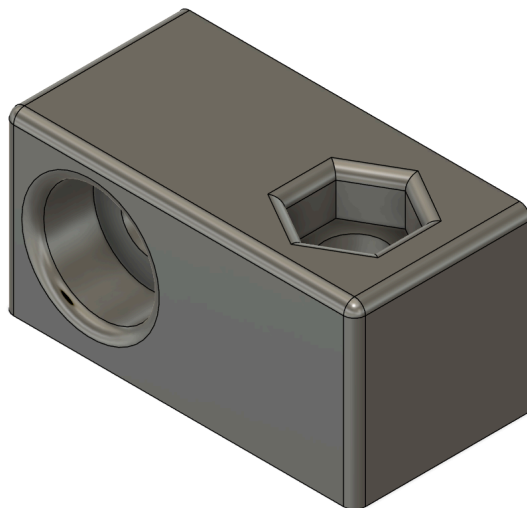
**Supporto
Telaio - Plexiglass**



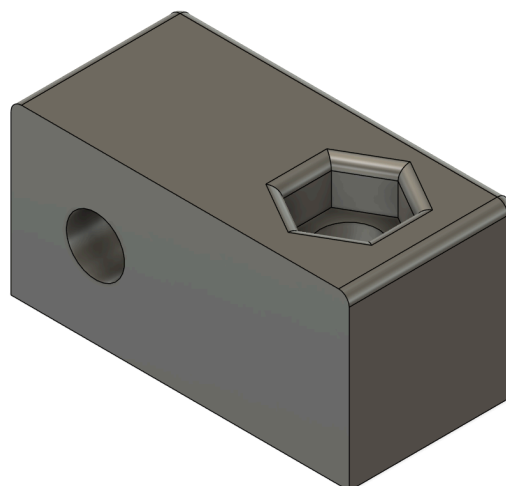
**Supporto
Telaio - Plexiglass**



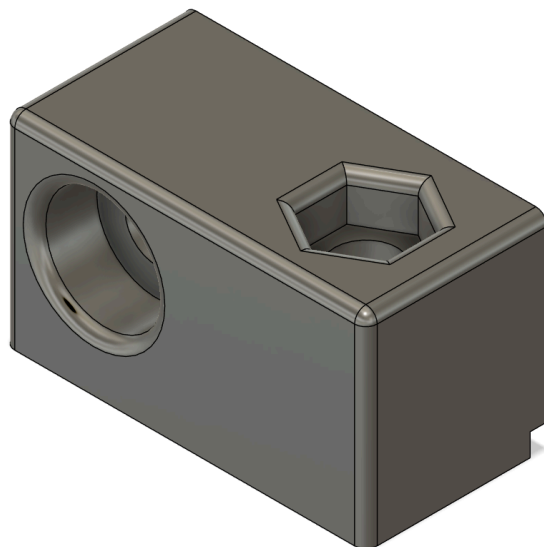
**Supporto
Telaio - Plexiglass**



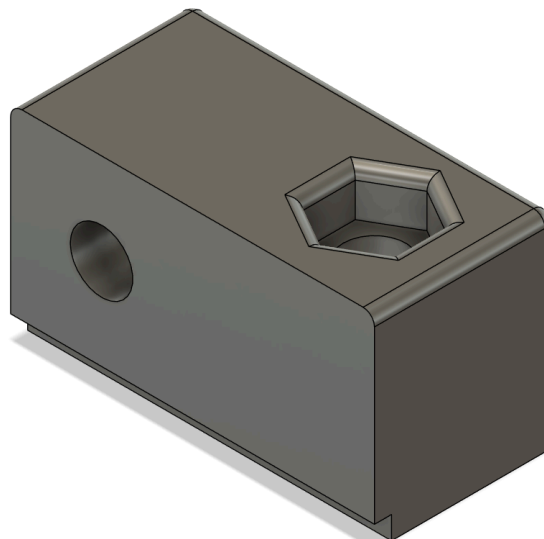
**Supporto
Telaio - Plexiglass**



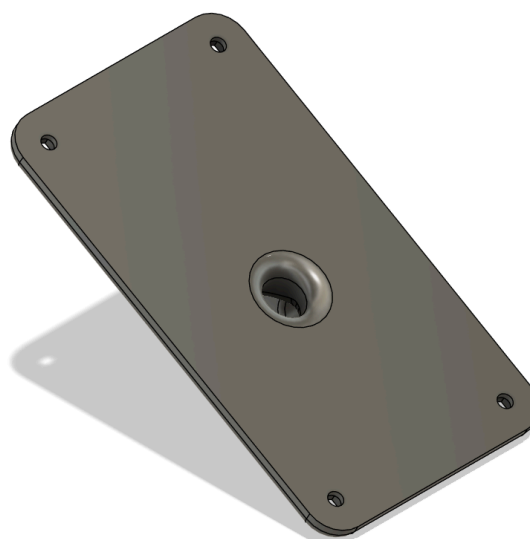
**Supporto
Telaio - Plexiglass**



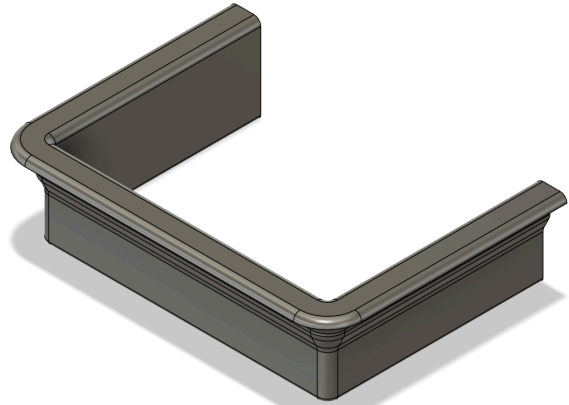
**Supporto
Telaio - Plexiglass**



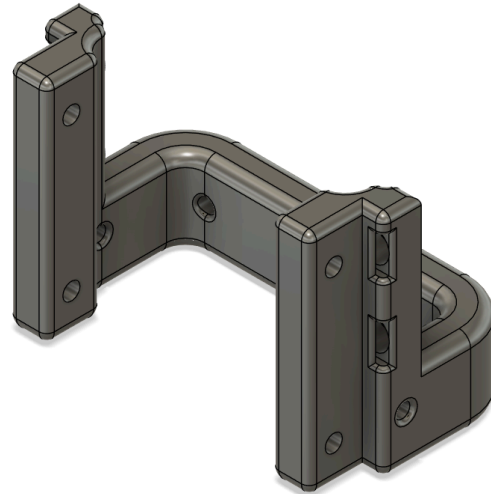
**Supporto
Telaio - Schermo**



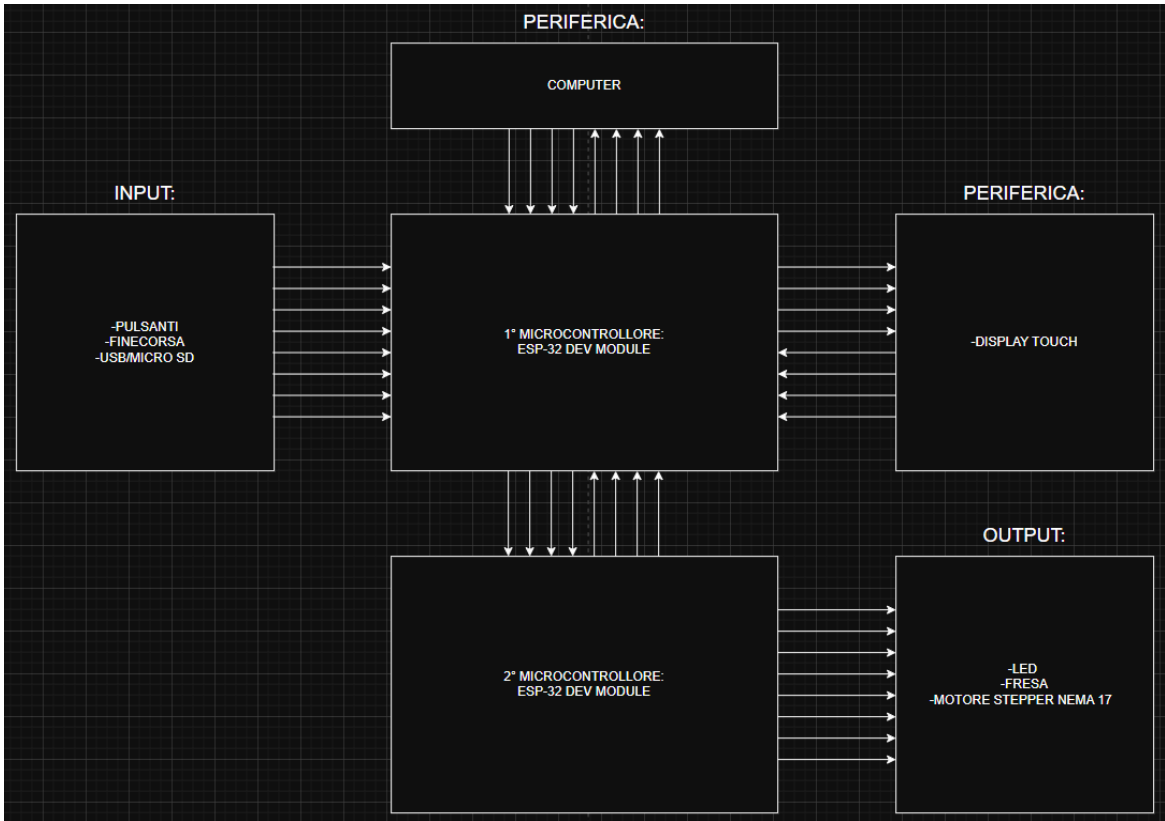
**Supporto
alimentatore - Plexiglass**



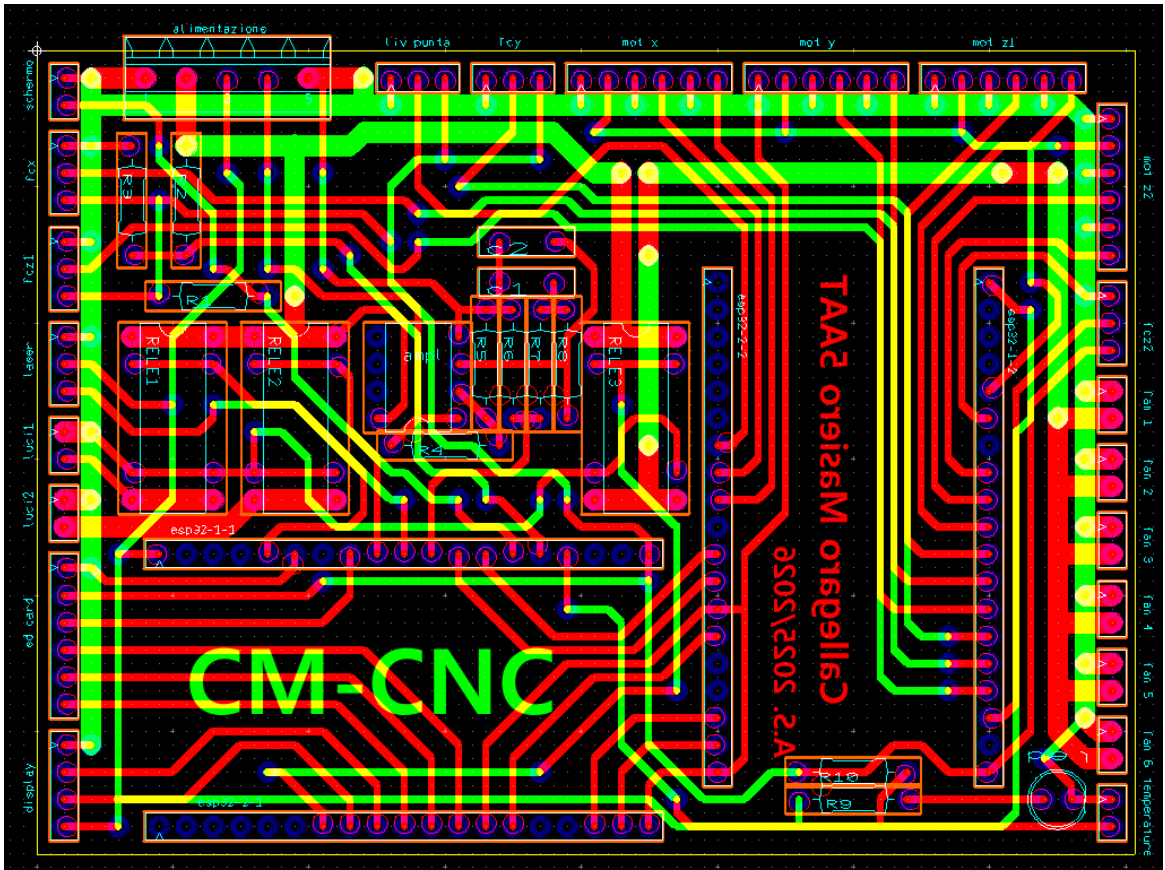
**Supporto
motore - Plexiglass**



SCHEMA A BLOCCHI:

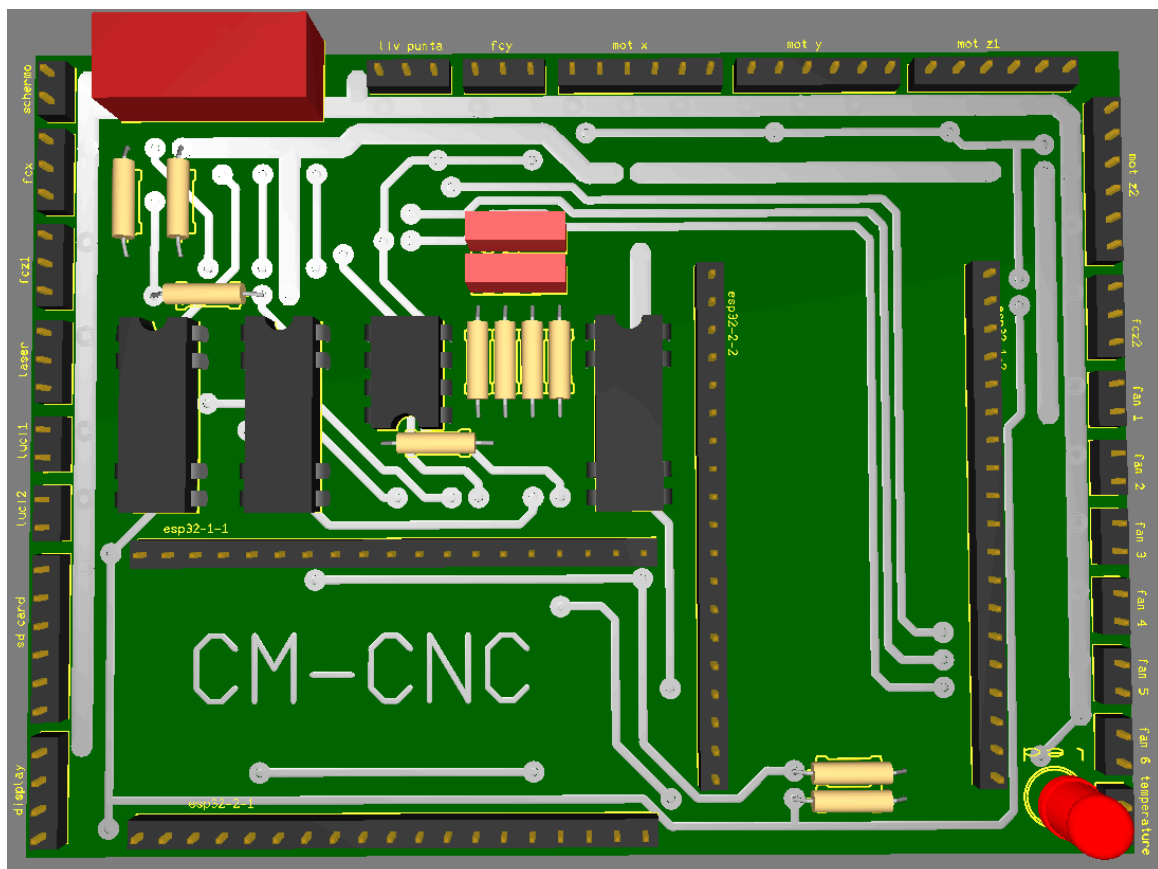
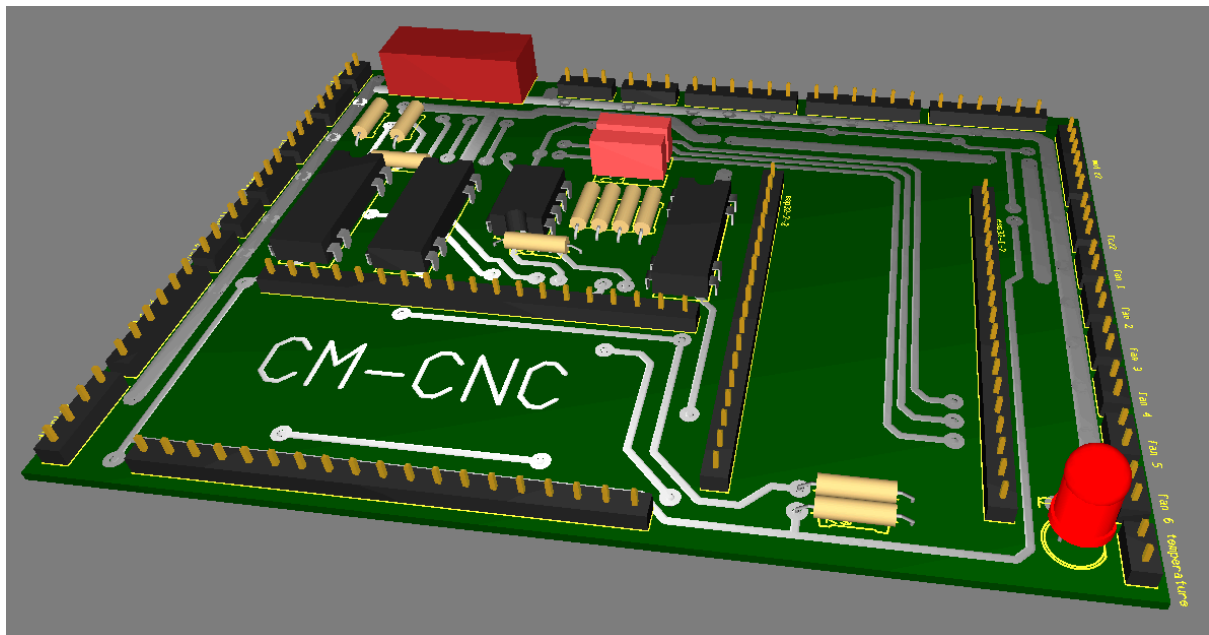


PCB Ultiboard:

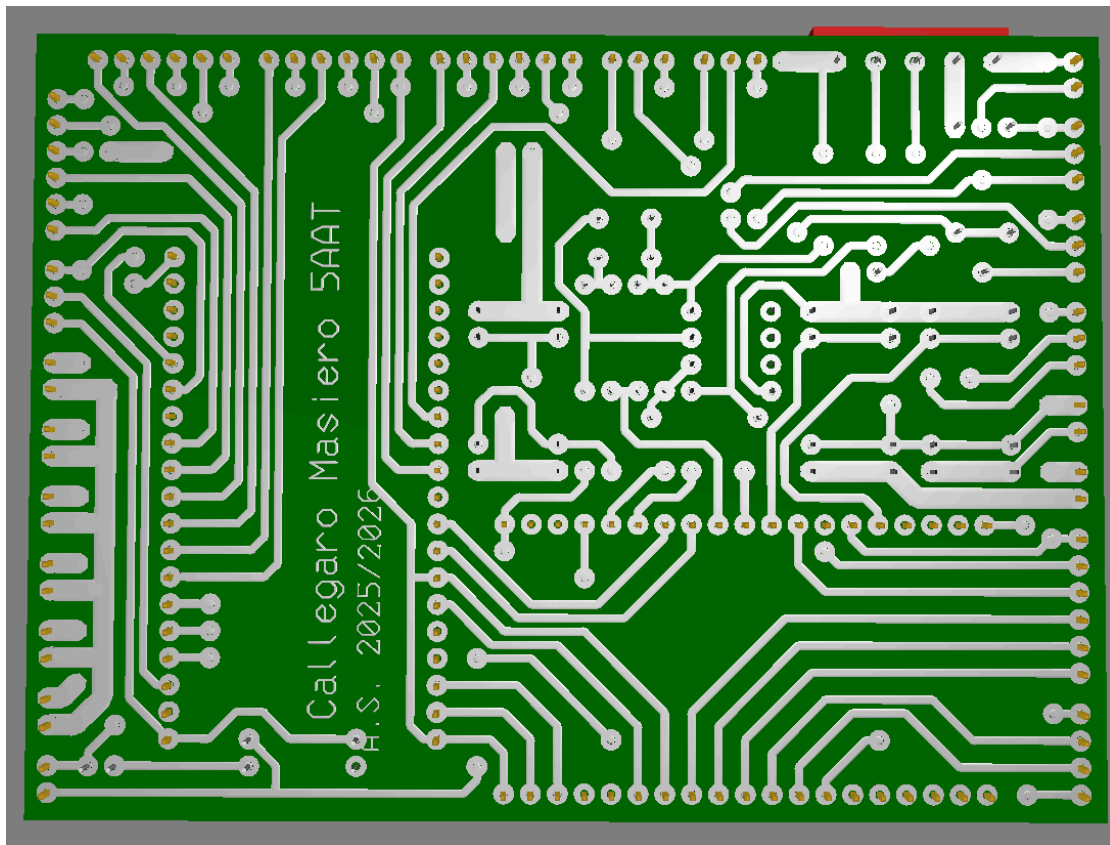


PCB Ultiboard vista anteprima 3D:

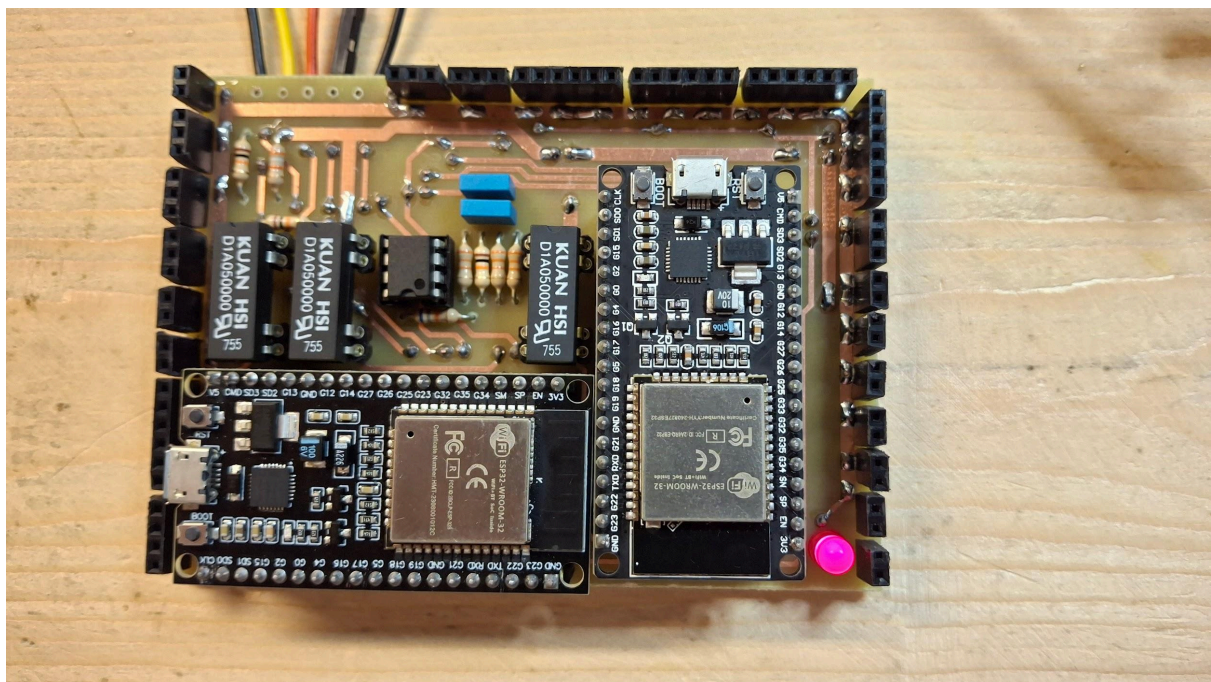
Copper top:



Copper bottom:



PCB ultimata:



LINK E FILE ESTERNI:

DIAGRAMMA DI GANTT:

[Gantt](#)

EXCEL MATERIALI ACQUISTATI:

[Excel Lista Materiali](#)

PRESENTAZIONE CANVA:

[PPTX-CNC-V1.0](#)

FILE DEL PROGETTO:

[github](#)